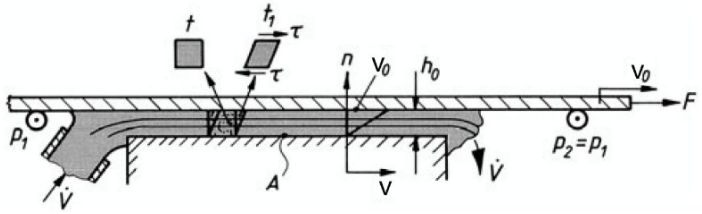
I. Viscous flow

Symbology

Symbol	Name	Units	Symbol	Name	Units
ν	Kinematic viscosity	m^2/s	η	Dynamic viscosity	Pa·s
τ	Shear stress	Pa	dn	Normal distance	m
du/dy	Rate of strain	[-]	e_{Diss}	Specific dissip. power	W/m^3

1. Flows with friction



$$F = \eta \frac{v_0 \cdot A}{h_0}$$

$$\tau = \frac{F}{A} = \eta \frac{v_0}{h_0} = \eta \frac{dv}{dn}$$

2. Viscosity

If $\eta = 0$, the fluid is called inviscid.

No slip condition: velocity of a particle closest to the wall has velocity equal to the wall velocity.

2.1 Reynolds Number

$$Re = \frac{\rho v L}{\eta} = \frac{vL}{\nu} \quad ; \quad \nu = \frac{\eta}{\rho}$$

2.2 Newtonian fluids

- Rate of deformation du/dy (velocity gradient) is linearly proportional to the shear stress τ
- The constant of proportionality is the viscosity (slope)
- $\eta = \dfrac{\tau}{du/dy} = \text{constant}$

2.3 Non-Newtonian fluids

Non-linear relation between shear stress and deformation rate due to non-constant viscosity.

$$\tau = \eta \left(\frac{\partial u}{\partial y}\right) \cdot \frac{\partial u}{\partial y}$$

2.3.1 Dilatant (shear thickening) fluids

Viscosity increases with increasing strain rate.

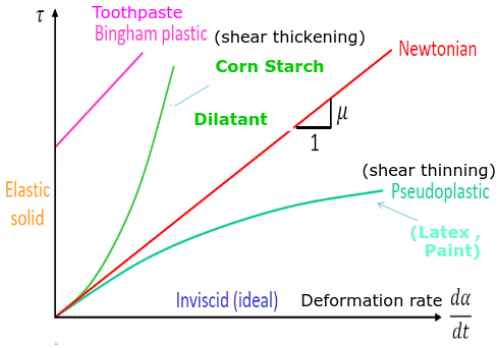
2.3.2 Plastic or pseudo-plastic (shear thinning) fluids

Viscosity decreases with increasing strain rate.

2.3.3 Bingham medium

Flow occurs only after a yield stress τ_0 is reached.

2.4 Graphical representation



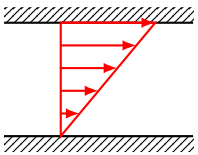
3. Flow profiles

3.1 Couette flow profiles (constant velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{v_p}{s}$$

$$e_{\text{Diss}} = \eta \left(\frac{\partial u(y)}{\partial y}\right)^2 = \eta \left(\frac{v_p}{s}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}(y)\, dV$$



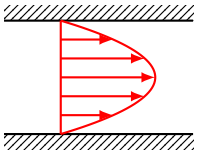
3.2 Poiseuille flow profiles (parabolic velocity gradient)

$$\frac{\partial u(y)}{\partial y} = \frac{4 v_{\text{max}}}{s}\left(1 - \frac{2y}{s}\right)$$

$$v(r) = ar^2 + br + c$$

$$e_{\text{Diss}} = \eta \frac{16 v_{\text{max}}^2}{s^2}\left(1 - \frac{2y}{s}\right)^2$$

$$P_{\text{Diss}} = \frac{16 \eta A v_{\text{max}}^2}{3s}$$



4. Laminar pipe flow

4.1 $p_1 > p_2$

$$\frac{\partial p}{\partial z} = \frac{p_2 - p_1}{L}$$

4.2 Constant linear flow profiles

$$\dot{V} = v_m A = v_m R^2 \pi$$

4.3 Parabolic flow profiles

$$\dot{V} = \int v(r)\, dA = \frac{v_{max} \cdot R^2 \pi}{2}$$

$$v(r) = v_{max}\left(1 - \frac{r^2}{R^2}\right)$$

$$\frac{dv}{dr} = -2 v_{max} \frac{r}{R^2}$$

$$P_{\text{Diss}} = \int_0^L \int_0^{2\pi} \int_0^R \eta \left(-2 v_{max} \frac{r^2}{R^2}\right) \cdot r\, dr\, d\varphi\, dx$$

$$P_{\text{Diss}} = 2\pi L \int_0^R \eta \left(-2 v_{max} \frac{r^2}{R^2}\right) \cdot r\, dr = 2\pi L \eta v_{max}^2 = 8\pi L \eta v_m^2$$

5. Flows in gaps and bearings

5.1 Assumptions in gaps and bearings

- Newtonian fluid with constant viscosity η
- Incompressible fluid
- Very low Reynolds number $Re < 2000$
- Laminar flow
- Balance of pressure and viscous forces
- No slip condition at the walls
- No acceleration (negligible)
- Pressure is uniform across the thickness $\partial p / \partial y \approx 0$

5.2 Equilibrium of forces

$$\sum F_x = 0 \Longrightarrow pA_p - (p + dp)A_p + \tau A_\tau + (\tau + d\tau)A_\tau = 0$$

$$A_p = dy \cdot b \quad ; \quad A_\tau = dx \cdot b \quad ; \quad \tau = \eta \frac{dv}{dy} \rightarrow \frac{d\tau}{dy}$$

$$\boxed{\eta \frac{d^2 v(x,y)}{dy^2} = \frac{dp(x)}{dx} = p'}$$

5.3 Basic velocity equation

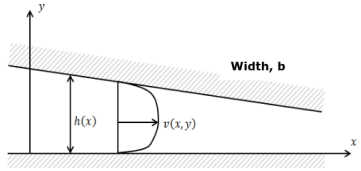
$$v(x,y) = \frac{1}{\eta} \cdot p' \cdot \frac{y^2}{2} + c(x)y + d(x)$$

5.3.1 Gap flow (walls not moving)

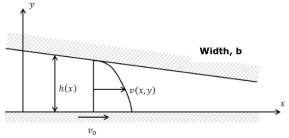
$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx}$$

$$d(x) = 0$$



5.3.2 Plain bearing / Creeping gap flow (one wall moving)



Assumption for $Re << 1$: $\eta \dfrac{d^2 v}{dy^2} = \dfrac{dp}{dx} = p'$

$$v(x,y) = \frac{p'(x)}{2\eta}\left[y^2 - h(x)y\right] + \frac{v_0}{h(x)}\left[h(x) - y\right]$$

$$c(x) = -\frac{h(x)}{2\eta} \cdot \frac{dp(x)}{dx} - \frac{v_0}{h(x)}$$

$$d(x) = v_0$$

For the simplest, parabolic velocity profile case:

$$h(x) = h_0 \quad ; \quad v_0 = 0 \quad ; \quad p' = \frac{dp(x)}{dx} = \frac{\Delta p}{\Delta L} = -\frac{12\eta v_m}{h_0^2}$$

$$v(y) = \frac{p'}{2\eta}\left(y^2 - h_0 y\right) \Longrightarrow \frac{\partial v}{\partial y} = 0 = \frac{p'}{2\eta}\left(2y - h_0\right)$$

$$y = \frac{h_0}{2} \Longrightarrow v_{max} = v\left(y = \frac{h_0}{2}\right) = -\frac{h_0^2 p'}{8\eta} = \frac{3}{2} v_m$$

$$v_m = -\frac{\Delta p \cdot h_0^2}{12\eta L}$$

$$\dot{V} = b \int_0^{h_0} v(y)\, dy = -b\frac{p' h_0^3}{12\eta} = \frac{h_0^2}{6\eta} b \Delta p$$

General sliding bearings:

$$\dot{V} = b \int_0^{h(x)} \frac{p'}{2}\eta\left(y^2 - h(x)y\right) + \frac{v_0}{h(x)}\left(h(x) - y\right)\, dy$$

$$\dot{V} = -b\frac{p' h_{(x)}^3}{12\eta} + b\frac{v_0 h(x)}{2} = ¢$$

$$p' = 6\eta v_0 \left(\frac{1}{h(x)^2} - \frac{\dot{V}}{b h(x)^3}\right)$$

5.3.3 General form of the volumetric flow rate

$$v(x,y) = \frac{1}{\eta} p' \frac{y^2}{2} + c(x) + d(x)$$

$$\dot{V} = b \int_0^{h(x)} v(x,y)\, dy = b\left[-\frac{h^3(x)}{6\eta} p' + c(x)\frac{h^2(x)}{2} + d(x)h(x)\right] = ¢$$

$$p' = \frac{6\eta}{h(x)^3}\left(\frac{\dot{V}}{b} - c(x)\frac{h(x)^2}{2} - d(x)h(x)\right)$$

5.4 Specific dissipation power

$$P = \tau A v = \eta \frac{\partial v}{\partial n} A v \quad ; \quad dP = dF dv$$

$$dF_\tau = d\tau dA = \eta \frac{dv}{dy} dx dz$$

$$e_{\text{Diss}} = \frac{dP_\tau}{dV} = \frac{\eta \frac{dv}{dy} dx dz dv}{dx dy dz} = \eta\left(\frac{dv}{dy}\right)^2 = \eta\left(\frac{\partial v}{\partial y}\right)^2$$

$$P_{\text{Diss}} = \int_V e_{\text{Diss}}\, dV = \iiint_V e_{\text{Diss}}\, dx dy dz$$

5.4.1 Pressure loss according to Bernoulli

$$p_1 - p_2 = \rho \frac{P_{\text{Diss}}}{\dot{m}} = \frac{P_{\text{Diss}}}{\dot{V}} = \frac{8\pi L \eta v_m}{R^2}$$